

Sample Exam

Medical Image Analysis

Student name: _____

Immatriculation number: _____

Exam Questions

Note: All the questions have equal weight.

1. In axial 2D acquisitions the slices lie in the transverse plane and are stacked along the head-to-toe direction. Voxel sizes are given as in-plane (two values) and through-plane (the slice thickness), and the field-of-view is image size $\times$ voxel size.

An axial acquisition uses in-plane voxel size $0.7 \times 0.7 \text{ mm}^2$ with 320 pixels in both directions, and through-plane voxel size 4.0 mm with 44 slices.

- (a) What is the field-of-view in the head-to-toe, left-to-right and anterior-to-posterior directions, respectively?

Solution: Answer: $176 \times 224 \times 224 \text{ mm}^3$.

Head-to-toe is the through-plane direction of an axial acquisition: $44 \times 4.0 = 176 \text{ mm}$.

Both in-plane directions give $320 \times 0.7 = 224 \text{ mm}$.

- (b) The anatomy that has to be covered extends 210 mm in the head-to-toe direction. Does this protocol cover it? If not, how many slices are needed at the same slice thickness, and by what factor does the acquisition time change if it is proportional to the number of slices?

Solution: Answer: **No** — the protocol covers only 176 mm of the required 210 mm. $\lceil 210/4.0 \rceil = \lceil 52.5 \rceil = \mathbf{53}$ slices are needed (53 slices cover 212 mm). The acquisition time grows by a factor $53/44 \approx \mathbf{1.20}$, i.e. about 20% longer.

Note that you must round *up*: 52 slices would cover only 208 mm and miss the anatomy.

- (c) Compute the anisotropy ratio of the voxels. This volume is then fed, without resampling, to a 3D convolutional network that was trained on 1.0 mm isotropic data. What goes wrong?

Solution: Answer: the anisotropy ratio is $4.0/0.7 \approx \mathbf{5.7}$ — the voxel is almost six times longer through-plane than in-plane.

A $3 \times 3 \times 3$ kernel therefore covers $2.1 \times 2.1 \times 12.0 \text{ mm}^3$ of real tissue instead of the $3 \times 3 \times 3 \text{ mm}^3$ it saw during training. The receptive field of every neuron is wildly asymmetric in physical space, so the spatial features the network learned simply do not correspond to the same physical structures any more. This is a domain shift the model was never trained for, and the fix is to resample to the training voxel size first.

Note that resampling does not create the missing information: upsampling the through-plane direction from 4.0 mm to 1.0 mm interpolates three samples out of every one that was really measured. The effective resolution in that direction is still 4 mm.

2. In atlas-based segmentation the atlas is registered to the image and then used inside the mixture model. Which of the two posteriors below is the one used by atlas-based segmentation, as opposed to plain EM with a Gaussian mixture model? π_n denotes a mixture weight estimated from the image itself. Please circle the correct choice.

$$\begin{aligned} p(c(x)=n \mid I(x)) &\propto \mathcal{N}(I(x) \mid \mu_n, \Sigma_n) \pi_n \\ p(c(x)=n \mid I(x)) &\propto \mathcal{N}(I(x) \mid \mu_n, \Sigma_n) p_{\text{atlas}}(c(x)=n) \end{aligned}$$

Solution: Answer: Right one.

The atlas prior $p_{\text{atlas}}(c(x)=n)$ *replaces* the free mixture weight π_n . Two things change as a result. The prior is now spatially varying — it says something different at every voxel, which is exactly the information plain EM lacks — and it no longer has to be estimated, so the class probabilities drop out of the optimisation entirely.

3. Registration problems are classified along two axes: whether the two images come from the *same or different subjects*, and whether they were acquired with the *same or different modalities*.
- (a) Which of the four combinations is the easiest and which is the hardest, and why?

Solution: Answer: easiest is **intra-subject, intra-modality**; hardest is **inter-subject, inter-modality**.

The same subject imaged twice with the same sequence shows the same anatomy with comparable intensities, so both the transformation and the similarity measure have an easy job. Different subjects imaged with different modalities combine two independent difficulties: the anatomies genuinely differ, and the intensities are not comparable. The relative difficulty of the two middle cases is a subjective judgement. Difficulty also depends on the anatomy — brain is considerably easier than abdomen.

- (b) Each axis primarily determines one component of the registration algorithm. Which axis determines which, and why?

Solution: Answer: the **subject** axis determines the **transformation model**; the **modality** axis determines the **loss function**.

Same subject means the same anatomy is present in both images, so a rigid or affine model is often enough — what changed is position, not shape. Different subjects have genuinely different anatomy, with no exact correspondence and possibly pathology, so a deformable model is required.

The modality determines whether intensities are comparable. Within a modality, SSD is appropriate. Across modalities it fails outright, and you need a statistical metric that is invariant to the intensity relationship — normalised cross-correlation for a linear relationship, mutual information for an arbitrary one.

4. Please indicate whether the statement is true or false by circling the correct choice in each of the following.
- (a) **TRUE** False The 3×3 mean filter is separable, $S_3 = \frac{1}{9}[1, 1, 1] * [1, 1, 1]^T$, which makes it cheaper to apply than a general 3×3 kernel.
 - (b) True **FALSE** A Gaussian has finite support, so a 3×3 structural element represents a Gaussian filter exactly.
 - (c) **TRUE** False Binomial filters are built by iterated convolution of $[1, 1]$ and have better properties in the frequency domain than mean filtering.
 - (d) True **FALSE** Increasing σ of a Gaussian filter smooths more, but the size of structural element required is independent of σ .

Solution: (a) TRUE — a separable 2D kernel can be applied as two 1D passes, which costs $O(k)$ per pixel instead of $O(k^2)$. The Gaussian and binomial kernels are separable as well.

(b) FALSE — the Gaussian has *infinite* support. Any finite structural element is only an approximation of it, truncated at some distance from the centre.

(c) TRUE — $S_3 = \frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$, obtained from successive convolutions of $[1, 1]$, and larger binomial kernels from successive convolutions of smaller ones. They are integer filters and behave better in the frequency domain than the box filter.

(d) FALSE — and (b) is the reason. Since the element only approximates the Gaussian by truncating it, a wider Gaussian needs a correspondingly larger element for the approximation to remain good. Using a 3×3 element with a large σ silently truncates most of the kernel.

5. A classification network takes an input image $x \in \mathbb{R}^{64 \times 64}$ with a single channel and maps it through hidden layers of 256, 128 and 64 units to 16 output classes. Consider only the *first* link, from the input to the 256-unit layer, and ignore biases throughout.

- (a) How many weights does this link have if it is fully connected?

Solution: Answer: $64 \times 64 \times 256 = 4096 \times 256 = \mathbf{1\,048\,576}$ weights.

- (b) How many weights does it have if it is instead a valid 5×5 convolution producing 256 output channels from the single input channel?

Solution: Answer: $5 \times 5 \times 1 \times 256 = \mathbf{6\,400}$ weights.

- (c) What is the ratio, and what property of the convolutional layer explains it?

Solution: Answer: $1\,048\,576 / 6\,400 = \mathbf{163.84}$, so the fully connected link has roughly 164 times more parameters.

The explanation is **weight sharing**: the same 5×5 kernel is applied at every position, so the parameter count depends only on the kernel size and the channel counts — and not at all on the image size. The fully connected count, by contrast, scales with the number of pixels. Doubling the image to 128×128 would quadruple the fully connected count to over four million while leaving the convolutional count at exactly 6 400.

Note also that the convolutional layer *lifts* the data to 256 channels at every spatial position rather than compressing the whole image into 256 numbers, so it retains far more information despite using far fewer parameters.

6. In atlas-based segmentation the model is $p(I(x) \mid \theta) = \sum_n \mathcal{N}(I(x) \mid \mu_n, \Sigma_n) p_{\text{atlas}}(c(x)=n)$ with $\theta = \{\mu_n, \Sigma_n\}_{n=1}^N$. Please explain why the intensity parameters μ_n and Σ_n still have to be estimated for every new MR image, but could in principle be fixed in advance for CT. Then state what else the atlas gives you compared with plain EM. You do not need to write down the equations.

Solution: Answer: *Why MRI needs them fitted.* MRI has **no absolute intensity scale**. The number stored in a voxel depends on the scanner, the coil, the sequence parameters and the reconstruction, so “the intensity of white matter” is not a fixed quantity — it differs between two scans of the same person on two machines. The Gaussian parameters therefore have to be re-estimated from each image.

Why CT could fix them. CT intensities are expressed in **Hounsfield units**, which are physically calibrated against water and air. A given tissue has a reproducible value across scanners, so μ_n and Σ_n can be tabulated in advance and all the parameters fixed.

What else the atlas buys. Two things beyond the accuracy gain. It supplies **spatially varying prior information** — at each voxel it says what is expected there, which is exactly what plain EM has no way of knowing. And it **removes the label ambiguity**: unsupervised EM returns clusters in an arbitrary order with no names attached, whereas with an atlas component n is grey matter. You know what you asked for, and there is no randomness left in which component is which.

7. Which of the following statements about interpolation is **inaccurate**?
- ☐ (A) Bilinear interpolation can equivalently be written as the parametric surface $I(x_1, x_2) = a + bx_1 + cx_2 + dx_1x_2$, whose four unknowns are fixed by the intensities at the four surrounding pixels.
 - ☐ (B) The trilinear extension adds the remaining monomials to give eight unknowns, fixed by the eight corner voxels of the cell containing the query point.
 - ☐ (C) Bicubic interpolation has 16 unknowns, determined by the four corner intensities together with their first and mixed derivatives — and since those derivatives must themselves be approximated from further pixels, bicubic effectively uses more than four samples.
 - ☐ (D) Because bilinear interpolation is a weighted average of the surrounding values, applying it repeatedly during an iterative registration leaves the effective spatial resolution unchanged.

Solution: Answer: D.

Averaging is exactly what costs you resolution. Bilinear interpolation smooths the image slightly every time it is applied, and during an optimisation loop it is applied thousands of

times, so the blurring compounds. This is one reason the standard strategy is bilinear *during* the optimisation, where speed matters, and bicubic for the *final* resampling, where quality matters and the cost is paid once.

8. Which of the following statements about the loss functions used for image restoration is **incorrect**?
- ☐ (A) MAE, MSE, NMSE, PSNR and SSIM are all computed directly between the prediction and the ground-truth image.
 - ☐ (B) SSIM compares many local patches of the two images and combines three components — luminance, contrast and structure.
 - ☐ (C) A perceptual loss passes both images through a previously trained network and compares intermediate features; that feature network must have been trained on the same task and the same modality as the restoration model, otherwise the comparison is meaningless.
 - ☐ (D) An adversarial loss is a distributional distance defined over *sets* of samples rather than between two paired images.

Solution: Answer: C.

The feature network does not have to match the task or the modality. The lecture explicitly allows a well-established network trained on *natural* images — VGG is the standard choice — or a network trained on CT images for some entirely different task. What is being borrowed is a general-purpose notion of what makes two images look different in terms of contextual content rather than pixel values, and that transfers surprisingly well across domains.

The motivation for all of this is that MSE-type losses produce visibly over-smoothed output; combining a perceptual loss with an adversarial one gave the most realistic denoised low-dose CT in the referenced work.

9. A semi-supervised auxiliary loss enforces consistency under spatial transformations. Two transformations ϕ_1, ϕ_2 are drawn per image and the loss is $\mathcal{L}(\phi_2 \circ \phi_1^{-1} \circ f(\phi_1 \circ x; \theta), f(\phi_2 \circ x; \theta))$. Which of the following statements is **incorrect**?
- ☐ (A) The loss requires no labels, so it can be evaluated on unlabelled images.
 - ☐ (B) It encodes the principle that if $x \Leftrightarrow y$ then $\phi \circ x \Leftrightarrow \phi \circ y$, which is the same principle that underlies data augmentation.
 - ☐ (C) The inverse ϕ_1^{-1} appears so that the two predictions are brought into a common spatial frame; without it they would sit in different frames and could not be compared pixel by pixel.
 - ☐ (D) Because the loss is defined on unlabelled data, it replaces the supervised loss rather than being added to it.

Solution: Answer: D.

It is an *auxiliary* loss: the total cost is the supervised loss over the labelled examples *plus* the consistency term, and the consistency term is summed over labelled and unlabelled examples alike. Dropping the supervised loss would remove the only thing that ties the network's output to the actual task — a network that predicts a constant map everywhere is perfectly consistent under any transformation, and perfectly useless.

10. When the network parameters are sampled from their posterior rather than optimised to a point estimate, the sampled parameters can perform noticeably worse on the task than the MAP estimate. What is the explanation given in the lecture?
- ☐ (A) In high dimensions the posterior is always unimodal, so every sample collapses onto the mean and no diversity is possible.
 - ☐ (B) For an isotropic Gaussian in d dimensions the expected distance of a sample from the mean grows like $\sqrt{d}$, so samples concentrate away from the mode — and since the likelihood $p(D \mid \theta, M)$ is the task loss, moving away from the mode means moving away from well-performing parameters.
 - ☐ (C) High-dimensional posteriors cannot be sampled at all, so any apparent sample is an artefact of the random number generator.
 - ☐ (D) In high dimensions the task loss becomes independent of the parameters, so performance is essentially random.
 - ☐ (E) Sampling error decreases with dimension, so the effect disappears entirely for large networks.

Solution: Answer: B.

This is the “typical set” phenomenon and it is thoroughly counter-intuitive: in high dimensions almost none of the probability mass sits near the mode, even though the mode is the single most probable point. Samples live in a thin shell at distance $\approx \sqrt{d}$ from the mean.

The consequence for uncertainty estimation is uncomfortable. Because $\log p(D \mid \theta, M)$ is precisely the training objective, a parameter vector drawn honestly from the posterior can sit in a region of poor task performance — so a faithful posterior sample and a well-performing network are not the same thing. Bounded sampling schemes have been proposed to mitigate this, but they do not fully solve it.

11. A Vision Transformer is applied to images of size 224×224 with 3 channels. It uses a patch size of $P = 16$, an embedding dimension of $D = 768$, and 12 attention heads. Recall that the image is split into $N = HW/P^2$ patches, each flattened and mapped by a patch-embedding matrix $E \in \mathbb{R}^{(P^2C) \times D}$, and that a learnable class token is prepended to the sequence. Circle True or False:
- (a) **TRUE** False The number of patch tokens is $N = 196$.
 - (b) **TRUE** False The patch-embedding matrix E is 768×768 and therefore contains 589 824 weights.
 - (c) True **FALSE** After the class token is added, the sequence entering the encoder still has length 196, because the class token takes the place of one patch.

- (d) True **FALSE** The attention matrix QK^T of a single head has $N \times N$ entries, so its size grows linearly with the number of tokens.

Solution: (a) TRUE — $N = HW/P^2 = (224 \times 224)/16^2 = 50\,176/256 = 196$, i.e. a 14×14 grid of patches.

(b) TRUE — the flattened patch has $P^2C = 16^2 \times 3 = 256 \times 3 = 768$ entries, and it is mapped to $D = 768$, so E is $768 \times 768 = 589\,824$. (That the two dimensions coincide is a property of this particular configuration, not a general rule.)

(c) FALSE — the class token is *prepended*, not substituted: the sequence length becomes $196 + 1 = 197$. It carries no patch content of its own; it exists so that the final representation at that position can be read out and passed to the classification MLP.

(d) FALSE — an $N \times N$ matrix has N^2 entries, so the cost grows **quadratically** with sequence length. Here that is $197^2 = 38\,809$ attention scores per head, and 12 heads per layer. This quadratic scaling is the reason patching exists at all: attending over all $224 \times 224 = 50\,176$ pixels individually would need over 2.5 billion scores per head.

12. Which of the following statements on self-supervised learning are correct (multiple answers are possible)?

- ☐ (A) Contrastive learning needs labelled examples in order to decide which pairs of images should be treated as positive and which as negative.
- ☐ (B) A pre-text task is a task that requires only images; the network trained on it is then fine-tuned with very few labelled examples on the task you actually care about.
- ☐ (C) In the *local* contrastive loss the transformations must either preserve the patch correspondences or be undone before the corresponding patches are compared.
- ☐ (D) Foundation models in this setting are obtained by supervised training at very large scale on carefully curated labelled data sets.
- ☐ (E) The InfoNCE loss maximises the similarity between two views of the same image through its numerator, and minimises the similarity to the samples of the negative set through its denominator.

Solution: Answer: (B), (C) and (E) are correct.

(A) is wrong, and it is wrong about the whole point of the method. The labels in contrastive learning are *surrogate* labels manufactured from the data itself: two random transformations of the same image are declared “close”, everything else “far”. No human annotation is involved — which is precisely why the approach is useful when labels are scarce. Note also that the transformations are not restricted to geometric ones; intensity changes are used deliberately, so that the representation becomes invariant to them.

(D) inverts the defining property. Foundation models here are built by **self-supervised** training at very large scale — masked auto-encoding and variations of contrastive learning — on transformer architectures. Requiring curated labels at that scale is exactly what these methods exist to avoid.